

CYBER SECURITY INTERNSHIP

DATE: 04-09-2026

TASK 5

**CAPTURE AND ANALYZE NETWORK
TRAFFIC USING WIRESHARK**

TOOLS USED: WIRESHARK

1. INTRODUCTION

Network traffic consists of packets that are continuously exchanged between devices when using the internet or communicating over a network. Analysing these packets helps in understanding how different protocols are used for communication and how data travels between systems.

For this task, **Wireshark** was used to capture live network packets generated during normal internet activity and a connectivity test. The captured traffic was then analysed using protocol-based filters to identify different protocols and traffic types.

2. OBJECTIVE

Capture live network packets and identify basic protocols and traffic types.

3. TOOLS USED

Wireshark - It is a free and open-source network protocol analyzer used to capture and inspect network packets. Each captured packet contains valuable information such as source and destination addresses, protocols used, timestamps, packet size and payload details.

Command Prompt - To generate network traffic using the ping 8.8.8.8 command.

Web browser- To access Youtube

4. PROCEDURE

- a) **Select a network interface** - Opened Wireshark and selected the **Wi-Fi** network interface.
- b) **Start packet capture** - Started the capture to record live network packets.
- c) **Generate network traffic** - Opened YouTube and played a video. Also used Command Prompt to run ping 8.8.8.8 to generate ICMP traffic.
- d) **Stop the capture** - After about one minute, stopped the packet capture.
- e) **Apply filters** - Used filters such as dns, tcp, icmp and tls to identify different protocols in the captured traffic.
- f) **Inspect packets** - Selected individual packets to view their protocol and other basic packet details.

Saved the complete packet capture as **Network_capture .pcap** file

5. PACKETS IDENTIFIED

Protocol	Full Form	Purpose
DNS	Domain Name System	Resolves domain names into IP addresses.
TCP	Transmission Control Protocol	Provides reliable communication between devices.
TLS	Transport Layer Security	Provides encrypted communication for secure network connections.
ICMP	Internet Control Message Protocol	Used for network connectivity testing and diagnostics.

6. DETAILS OF PACKETS

Packet No.	Protocol	Source	Destination	Length	Important Details
322	DNS	Local IPv6 address	Local IPv6 address	203 bytes	DNS response for i.ytimg.com, UDP port 53 → 53758
14377	TLSv1.2	192.168.88.6	204.79.197.222	89 bytes	Application Data, TCP port 53493 → 443
14379	TCP	IPv6 address	IPv6 address	75 bytes	TCP Keep-Alive, port 19123 → 443, ACK
14414	ICMP	8.8.8.8	192.168.88.6	74 bytes	Echo (ping) Reply, TTL 115

7. LEARNINGS

This task provided practical insight into how network communication takes place at the packet level. By analyzing the captured traffic, I learned that a simple activity such as accessing YouTube involves multiple protocols working together, including DNS for domain resolution, TCP for reliable communication, TLS for encrypted data transfer and ICMP for connectivity testing.

Examining individual packets helped me understand how information such as source and destination IP addresses, ports, packet length and protocol-specific details can reveal how devices communicate across a network. Using filters made it easier to separate different types of traffic and focus on specific protocols, which demonstrated how packet analysis can be useful for identifying communication patterns, troubleshooting network issues and understanding network behaviour.

Thus, this task gave me practical experience with packet capture, protocol analysis, TCP/IP communication and network traffic filtering.